

A Polynomial Parametrization of

$$xyz + t^2 - 1 = 0$$

Abstract

We exhibit two polynomial maps over $\mathbb{Z}$ whose union parametrizes every rational solution of $xyz + t^2 - 1 = 0$. Integer parameter values yield an explicit three-parameter family of integer solutions. We also identify the inverse map on the nondegenerate locus, record its exact integrality criterion, and explain the factorization that leads naturally to the construction.

1 The parametrization

Let

$$X = \{(x, y, z, t) : xyz + t^2 - 1 = 0\}.$$

For each $\varepsilon \in \{1, -1\}$, define

$$\Phi_\varepsilon(a, b, c) = (a, b, c(2 - abc), \varepsilon(1 - abc)).$$

Every coordinate of Φ_ε is a polynomial in a, b, c with integer coefficients.

Theorem 1. *The following statements hold.*

1. *For every commutative ring R and every $(a, b, c) \in R^3$, the point $\Phi_\varepsilon(a, b, c)$ satisfies $xyz + t^2 - 1 = 0$.*
2. *The two polynomial maps are jointly exhaustive over the rationals:*

$$X(\mathbb{Q}) = \Phi_1(\mathbb{Q}^3) \cup \Phi_{-1}(\mathbb{Q}^3).$$

3. *On the open locus $xy \neq 0$, the map Φ_1 has the inverse*

$$(x, y, z, t) \mapsto \left(x, y, \frac{1-t}{xy}\right).$$

Consequently, X is a rational affine threefold; more precisely, Φ_1 identifies the open set $ab \neq 0$ in $\mathbb{A}_{\mathbb{Q}}^3$ with the open set $xy \neq 0$ in X .

Proof. Put $s = abc$. For $(x, y, z, t) = \Phi_\varepsilon(a, b, c)$, one has

$$xyz = abc(2 - abc) = s(2 - s),$$

and, since $\varepsilon^2 = 1$,

$$t^2 - 1 = (1 - s)^2 - 1 = s^2 - 2s = -s(2 - s).$$

Their sum is zero. This proves the first assertion as a polynomial identity over $\mathbb{Z}$, and hence over every commutative ring.

Now let $(x, y, z, t) \in X(\mathbb{Q})$. Suppose first that $xy \neq 0$, and set

$$a = x, \quad b = y, \quad c = \frac{1-t}{xy}.$$

Then $abc = 1 - t$, so the fourth coordinate of $\Phi_1(a, b, c)$ is t . Moreover,

$$\begin{aligned} c(2 - abc) &= \frac{1-t}{xy}(2 - (1-t)) \\ &= \frac{(1-t)(1+t)}{xy} \\ &= \frac{1-t^2}{xy} = z, \end{aligned}$$

where the final equality uses $xyz = 1 - t^2$. Thus $(x, y, z, t) = \Phi_1(a, b, c)$.

If $xy = 0$, then the defining equation gives $t^2 = 1$. When $t = 1$,

$$(x, y, z, t) = \Phi_1\left(x, y, \frac{z}{2}\right),$$

and when $t = -1$,

$$(x, y, z, t) = \Phi_{-1}\left(x, y, \frac{z}{2}\right).$$

This proves the second assertion. On $xy \neq 0$, the preceding formula for c is forced by $t = 1 - abc$, so it is the inverse of Φ_1 . The final statement follows. $\square$

Corollary 2. *For arbitrary $a, b, c \in \mathbb{Z}$,*

$$\boxed{(x, y, z, t) = (a, b, c(2 - abc), \pm(1 - abc))}$$

is an integer solution. In particular, the equation has infinitely many integer solutions.

Proof. The first statement is Theorem 1(1) with $R = \mathbb{Z}$. For an explicit one-parameter subfamily, take $a = b = 1$ and $c = n$. This gives

$$(x, y, z, t) = (1, 1, n(2 - n), 1 - n), \quad n \in \mathbb{Z}.$$

Different values of n give different fourth coordinates, and hence different solutions. $\square$

2 How the construction is found

The equation is most naturally rewritten as

$$xyz = 1 - t^2 = (1 - t)(1 + t).$$

The right-hand side already consists of two factors whose difference is exactly 2. To introduce three free parameters while preserving this factorization, prescribe one of those factors to be their product:

$$1 - t = abc.$$

This immediately forces

$$t = 1 - abc, \quad 1 + t = 2 - abc.$$

Therefore

$$1 - t^2 = (1 - t)(1 + t) = abc(2 - abc).$$

The product on the right can now be distributed among x, y, z by taking

$$x = a, \quad y = b, \quad z = c(2 - abc).$$

This gives the displayed polynomial parametrization without any division. Choosing instead the sign-reversed fourth coordinate produces the second chart Φ_{-1} .

The essential idea is therefore simple but structural: force one linear factor of $1 - t^2$ to absorb the product of the free parameters; the other linear factor then remains polynomial because the two factors differ by 2.

3 Rational completeness versus integral completeness

Theorem 1 is exhaustive when the parameters are allowed to be rational. With integer parameters, the image is still a large three-parameter family, but it does not contain every integer point. The precise obstruction on the nondegenerate locus is divisibility.

Proposition 3. *Let $(x, y, z, t) \in X(\mathbb{Z})$ with $xy \neq 0$.*

1. *It has an integral preimage under Φ_1 if and only if $xy \mid 1 - t$. In that case the preimage is uniquely*

$$\left(x, y, \frac{1 - t}{xy}\right).$$

2. *It has an integral preimage under Φ_{-1} if and only if $xy \mid 1 + t$. In that case the preimage is uniquely*

$$\left(x, y, \frac{1 + t}{xy}\right).$$

Proof. For Φ_1 , the first two coordinates force $a = x$ and $b = y$, while the fourth coordinate forces $c = (1 - t)/(xy)$. Thus an integral preimage exists exactly when this quotient is integral, and it is then unique. The argument for Φ_{-1} , whose fourth coordinate is $abc - 1$, is identical. $\square$

Remark 4. The distinction is genuine. For example,

$$(30, 42, -4, 71) \in X(\mathbb{Z}),$$

because

$$30 \cdot 42 \cdot (-4) + 71^2 - 1 = -5040 + 5041 - 1 = 0.$$

The pairwise products of the first three coordinates are 1260, -120 , -168 , and none divides either $1 - 71 = -70$ or $1 + 71 = 72$. Hence this point is not obtained from integer parameters in either chart, even after permuting x, y, z . Thus the formulas give a complete polynomial parametrization over $\mathbb{Q}$ and an explicit infinite polynomial family over $\mathbb{Z}$, but not a claimed complete parametrization of all integral points by unrestricted integer parameters.